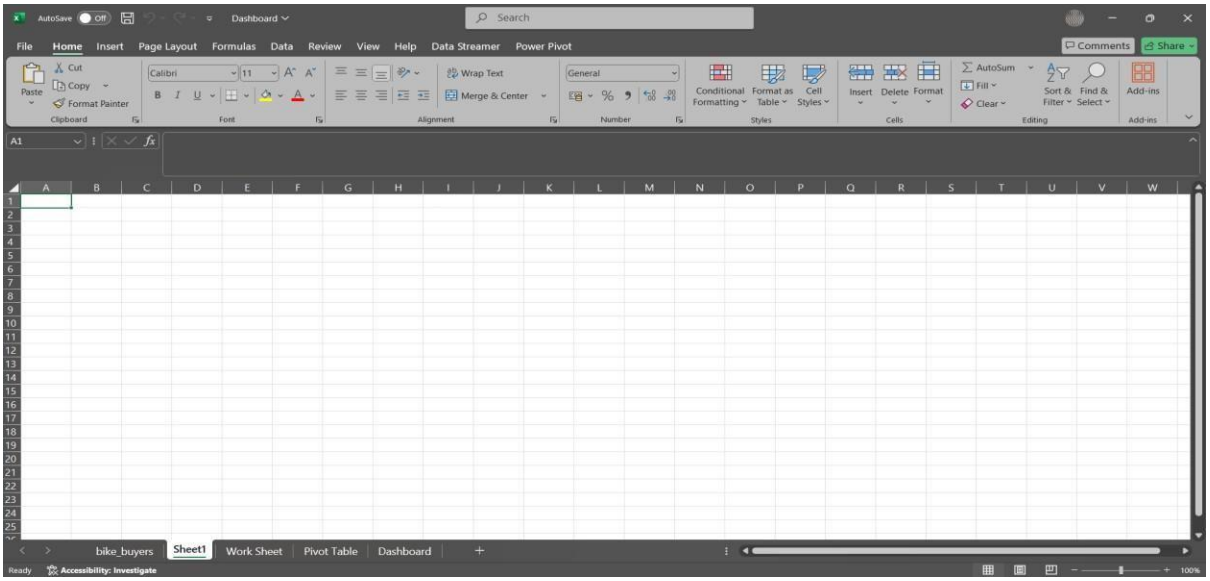
EX.NO: 12	DATA ANALYSIS AND VISUALIZATION USING ADVANCED EXCEL
DATE:	

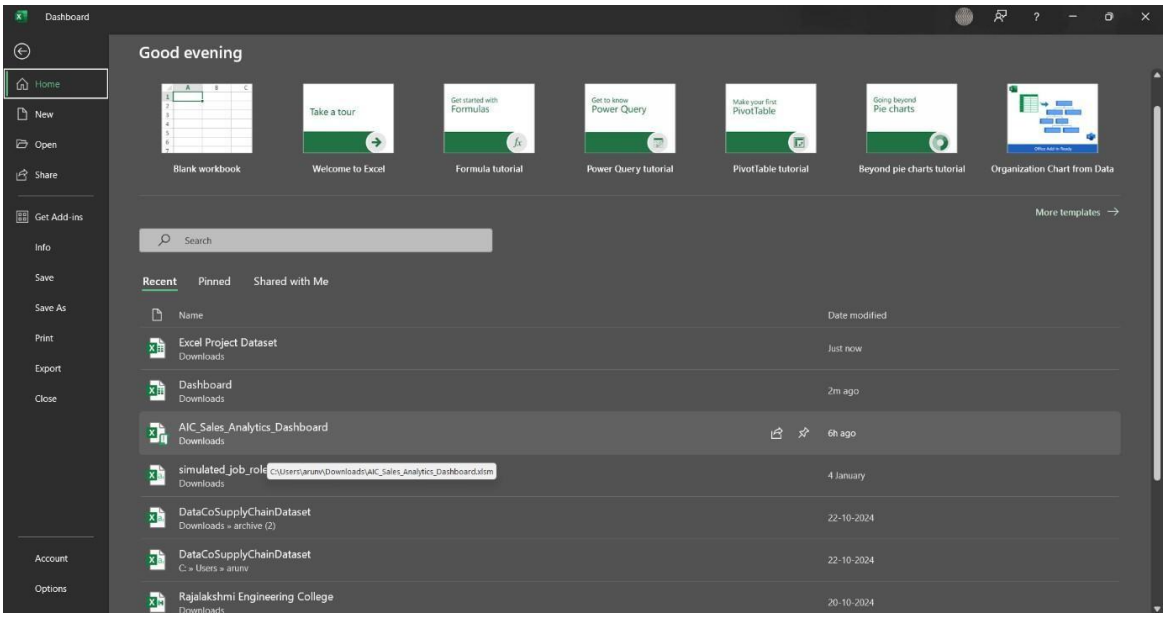
AIM:

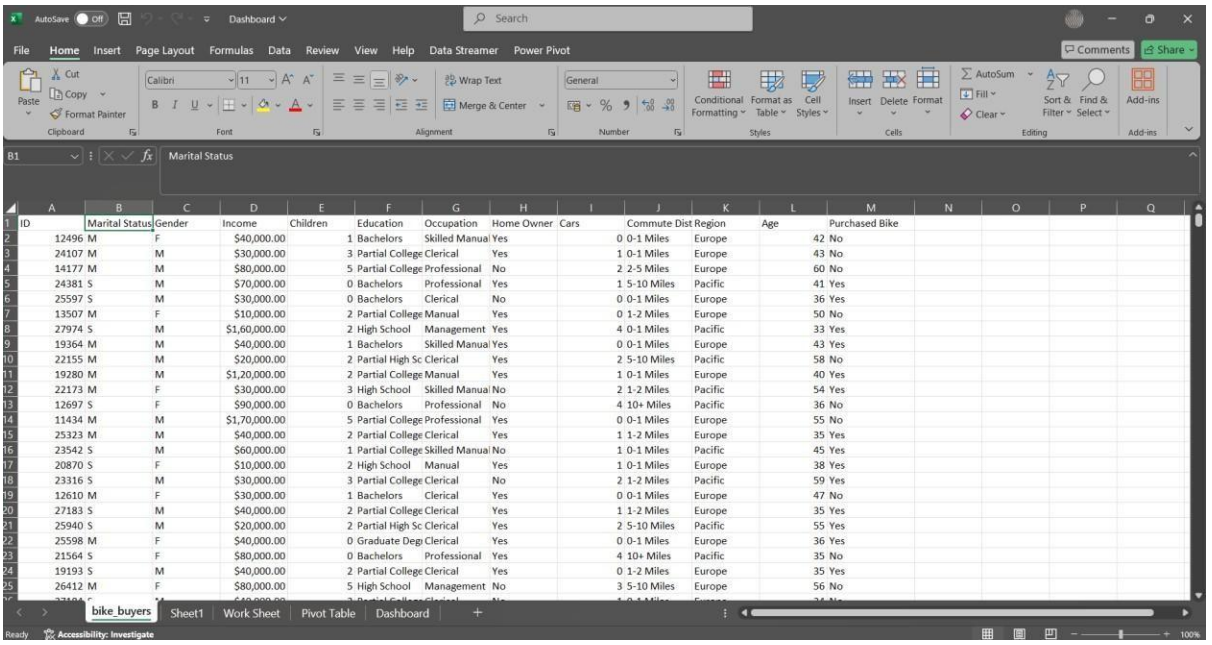
To implement the Data Analysis and Visualization using Advanced Excel.

PROCEDURE:

Step 1: Open the blank excel and import the Dataset into Microsoft Excel 2021.

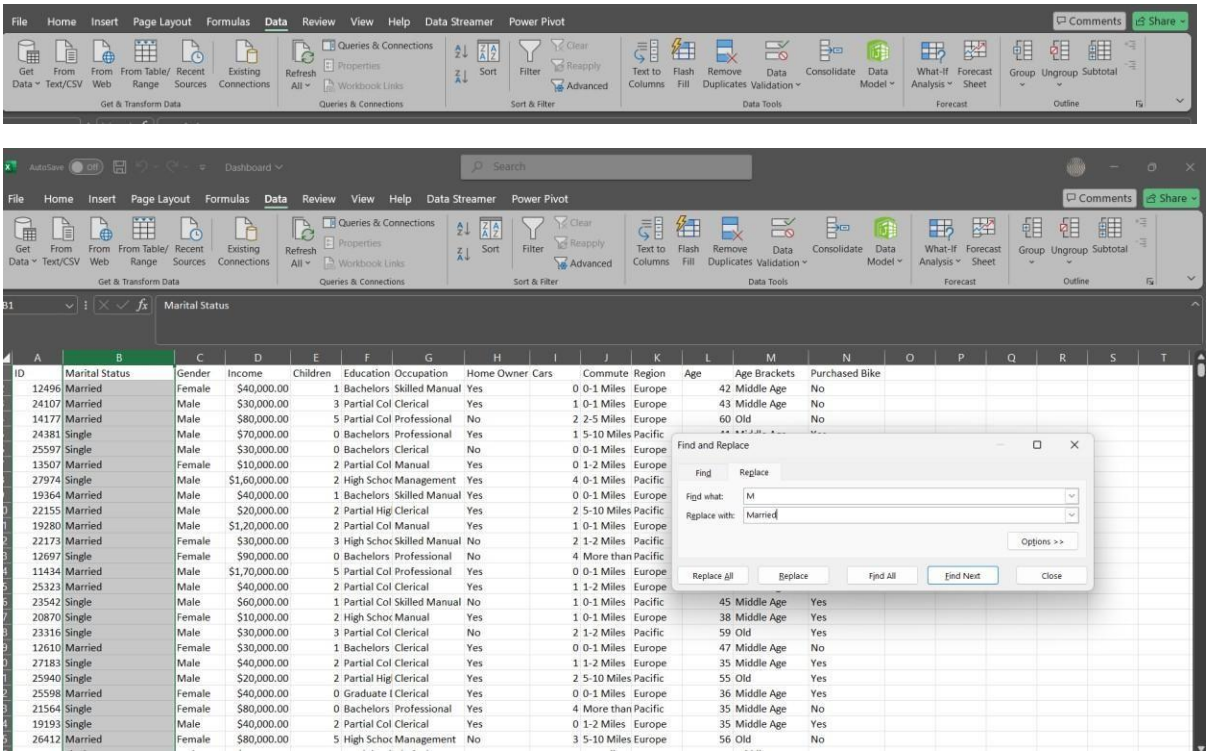






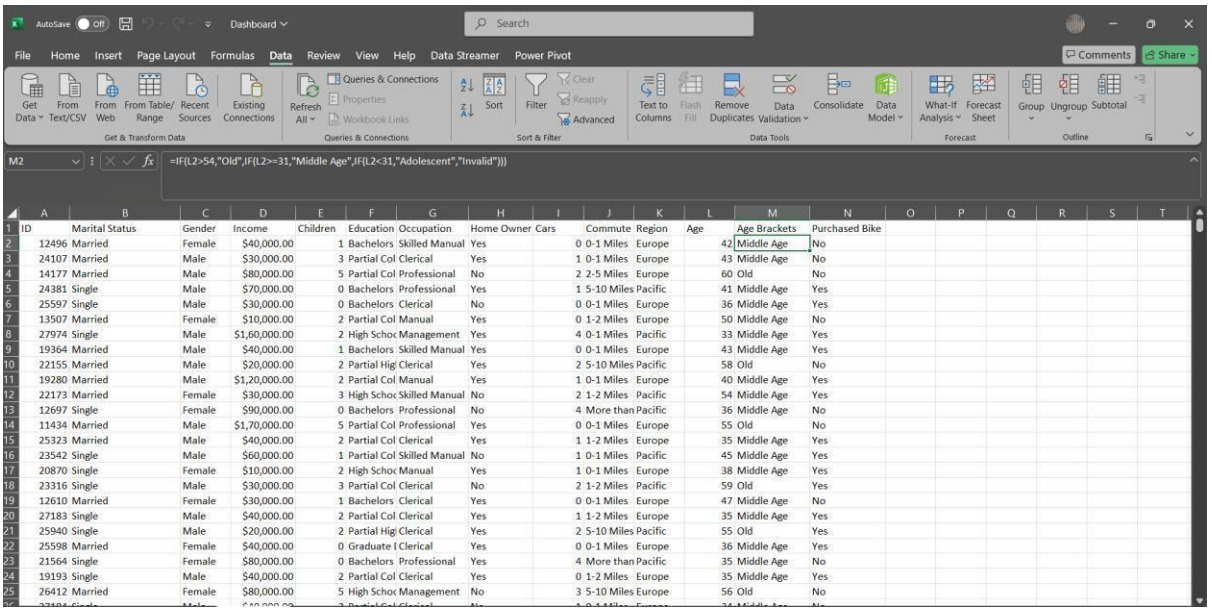
ID	Marital Status	Gender	Income	Children	Education	Occupation	Home Owner	Cars	Commute Dist	Region	Age	Purchased Bike
12496	M	F	$40,000.00	1	Bachelors	Skilled Manual	Yes	0	0-1 Miles	Europe	42	No
24107	M	M	$30,000.00	3	Partial College	Clerical	Yes	1	0-1 Miles	Europe	43	No
14177	M	M	$80,000.00	5	Partial College	Professional	No	2	2-5 Miles	Europe	60	No
24381	S	M	$70,000.00	0	Bachelors	Professional	Yes	1	5-10 Miles	Pacific	41	Yes
25597	S	M	$30,000.00	0	Bachelors	Clerical	No	0	0-1 Miles	Europe	36	Yes
13507	M	F	$10,000.00	2	Partial College	Manual	Yes	0	1-2 Miles	Europe	50	No
27974	S	M	$1,60,000.00	2	High School	Management	Yes	4	0-1 Miles	Pacific	33	Yes
19364	M	M	$40,000.00	1	Bachelors	Skilled Manual	Yes	0	0-1 Miles	Europe	43	Yes
22155	M	M	$20,000.00	2	Partial High Sc	Clerical	Yes	2	5-10 Miles	Pacific	58	No
19280	M	M	$1,20,000.00	2	Partial College	Manual	Yes	1	0-1 Miles	Europe	40	Yes
22173	M	F	$30,000.00	3	High School	Skilled Manual	No	2	1-2 Miles	Pacific	54	Yes
12697	S	F	$90,000.00	0	Bachelors	Professional	No	4	10+ Miles	Pacific	36	No
11434	M	M	$1,70,000.00	5	Partial College	Professional	Yes	0	0-1 Miles	Europe	55	No
25323	M	M	$40,000.00	2	Partial College	Clerical	Yes	1	1-2 Miles	Europe	35	Yes
23542	S	M	$60,000.00	1	Partial College	Skilled Manual	No	1	0-1 Miles	Pacific	45	Yes
20870	S	F	$10,000.00	2	High School	Manual	Yes	1	0-1 Miles	Europe	38	Yes
23316	S	M	$30,000.00	3	Partial College	Clerical	No	2	1-2 Miles	Pacific	59	Yes
12610	M	F	$30,000.00	1	Bachelors	Clerical	Yes	0	0-1 Miles	Europe	47	No
27183	S	M	$40,000.00	2	Partial College	Clerical	Yes	1	1-2 Miles	Europe	35	Yes
25940	S	M	$20,000.00	2	Partial High Sc	Clerical	Yes	2	5-10 Miles	Pacific	55	Yes
25598	M	F	$40,000.00	0	Graduate Degr	Clerical	Yes	0	0-1 Miles	Europe	36	Yes
21564	S	F	$80,000.00	0	Bachelors	Professional	Yes	4	10+ Miles	Pacific	35	No
19193	S	M	$40,000.00	2	Partial College	Clerical	Yes	0	1-2 Miles	Europe	35	Yes
26412	M	F	$80,000.00	5	High School	Management	No	3	5-10 Miles	Europe	56	No

Step 2: Cleaning and pre-processing the dataset for the better visualization. Using the key Data you can remove the duplicates in the entries. To know clearly about the dataset for the user making changes in row 2 & 3.



ID	Marital Status	Gender	Income	Children	Education	Occupation	Home Owner	Cars	Commute	Region	Age	Age Brackets	Purchased Bike
12496	Married	Female	$40,000.00	1	Bachelors	Skilled Manual	Yes	0	0-1 Miles	Europe	42	Middle Age	No
24107	Married	Male	$30,000.00	3	Partial Col	Clerical	Yes	1	0-1 Miles	Europe	43	Middle Age	No
14177	Married	Male	$80,000.00	5	Partial Col	Professional	No	2	2-5 Miles	Europe	60	Old	No
24381	Single	Male	$70,000.00	0	Bachelors	Professional	Yes	1	5-10 Miles	Pacific			
25597	Single	Male	$30,000.00	0	Bachelors	Clerical	No	0	0-1 Miles	Europe			
13507	Married	Female	$10,000.00	2	Partial Col	Manual	Yes	0	1-2 Miles	Europe			
27974	Single	Male	$1,60,000.00	2	High Schoc	Management	Yes	4	0-1 Miles	Pacific			
19364	Married	Male	$40,000.00	1	Bachelors	Skilled Manual	Yes	0	0-1 Miles	Europe			
22155	Married	Male	$20,000.00	2	Partial Hig	Clerical	Yes	2	5-10 Miles	Pacific			
19280	Married	Male	$1,20,000.00	2	Partial Col	Manual	Yes	1	0-1 Miles	Europe			
22173	Married	Female	$30,000.00	3	High Schoc	Skilled Manual	No	2	1-2 Miles	Pacific			
12697	Single	Female	$90,000.00	0	Bachelors	Professional	No	4	More than	Pacific			
11434	Married	Male	$1,70,000.00	5	Partial Col	Professional	Yes	0	0-1 Miles	Europe			
25323	Married	Male	$40,000.00	2	Partial Col	Clerical	Yes	1	1-2 Miles	Europe			
23542	Single	Male	$60,000.00	1	Partial Col	Skilled Manual	No	1	0-1 Miles	Pacific	45	Middle Age	Yes
20870	Single	Female	$10,000.00	2	High Schoc	Manual	Yes	1	0-1 Miles	Europe	38	Middle Age	Yes
23316	Single	Male	$30,000.00	3	Partial Col	Clerical	No	2	1-2 Miles	Pacific	59	Old	Yes
12610	Married	Female	$30,000.00	1	Bachelors	Clerical	Yes	0	0-1 Miles	Europe	47	Middle Age	No
27183	Single	Male	$40,000.00	2	Partial Col	Clerical	Yes	1	1-2 Miles	Europe	35	Middle Age	Yes
25940	Single	Male	$20,000.00	2	Partial Hig	Clerical	Yes	2	5-10 Miles	Pacific	55	Old	Yes
25598	Married	Female	$40,000.00	0	Graduate I	Clerical	Yes	0	0-1 Miles	Europe	36	Middle Age	Yes
21564	Single	Female	$80,000.00	0	Bachelors	Professional	Yes	4	More than	Pacific	35	Middle Age	No
19193	Single	Male	$40,000.00	2	Partial Col	Clerical	Yes	0	1-2 Miles	Europe	35	Middle Age	Yes
26412	Married	Female	$80,000.00	5	High Schoc	Management	No	3	5-10 Miles	Europe	56	Old	No

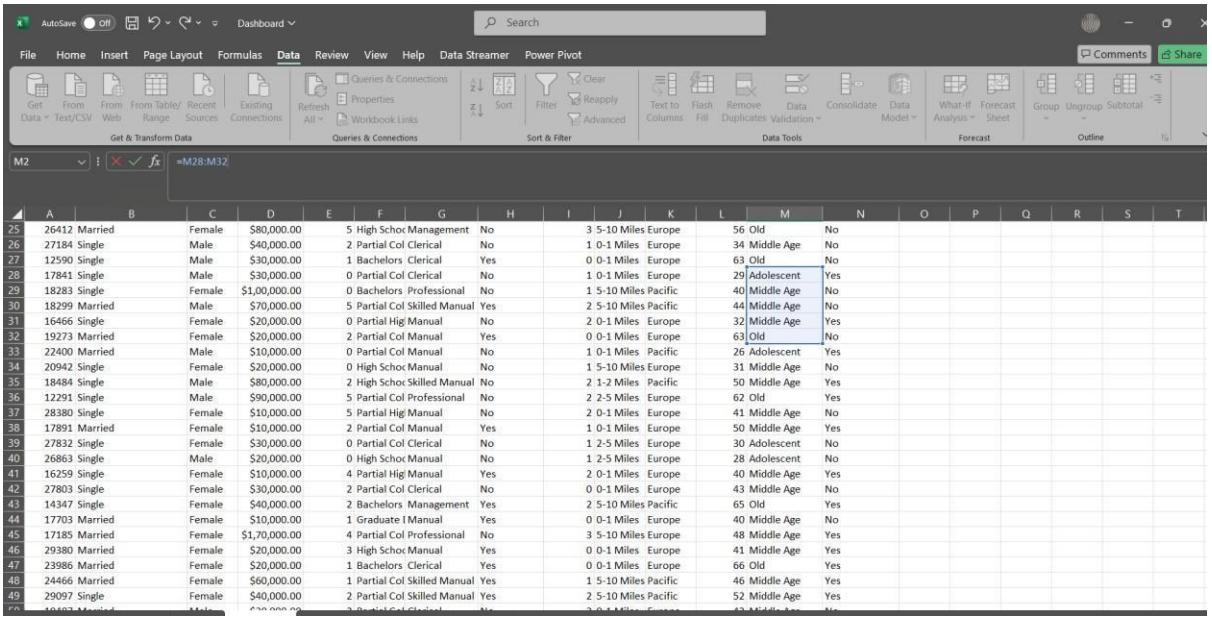
Step 3: For the Bike sales we need to capture the specific age group so include a row (Age Brackets) for the better view.



=IF(L2>54,"Old",IF(L2>=31,"Middle Age",IF(L2<31,"Adolescent","Invalid")))

ID	Marital Status	Gender	Income	Children	Education	Occupation	Home Owner	Cars	Commute	Region	Age	Age Brackets	Purchased Bike
12496	Married	Female	$40,000.00	1	Bachelors	Skilled Manual	Yes	0	0-1 Miles	Europe	42	Middle Age	No
24107	Married	Male	$30,000.00	3	Partial Col	Clerical	Yes	1	0-1 Miles	Europe	43	Middle Age	No
14177	Married	Male	$80,000.00	5	Partial Col	Professional	No	2	2-5 Miles	Europe	60	Old	No
24381	Single	Male	$70,000.00	0	Bachelors	Professional	Yes	1	5-10 Miles	Pacific	41	Middle Age	Yes
25597	Single	Male	$30,000.00	0	Bachelors	Clerical	No	0	0-1 Miles	Europe	36	Middle Age	Yes
13507	Married	Female	$10,000.00	2	Partial Col	Manual	Yes	0	1-2 Miles	Europe	50	Middle Age	No
27974	Single	Male	$1,60,000.00	2	High Schoo	Management	Yes	4	0-1 Miles	Pacific	33	Middle Age	Yes
19364	Married	Male	$40,000.00	1	Bachelors	Skilled Manual	Yes	0	0-1 Miles	Europe	43	Middle Age	Yes
22155	Married	Male	$20,000.00	2	Partial Hig	Clerical	Yes	2	5-10 Miles	Pacific	58	Old	No
19280	Married	Male	$1,20,000.00	2	Partial Col	Manual	Yes	1	0-1 Miles	Europe	40	Middle Age	Yes
22173	Married	Female	$30,000.00	3	High Schoo	Skilled Manual	No	2	1-2 Miles	Pacific	54	Middle Age	Yes
12697	Single	Female	$90,000.00	0	Bachelors	Professional	No	4	More than	Pacific	36	Middle Age	No
11434	Married	Male	$1,70,000.00	5	Partial Col	Professional	Yes	0	0-1 Miles	Europe	55	Old	No
25323	Married	Male	$40,000.00	2	Partial Col	Clerical	Yes	1	1-2 Miles	Europe	35	Middle Age	Yes
23542	Single	Male	$60,000.00	1	Partial Col	Skilled Manual	No	1	0-1 Miles	Pacific	45	Middle Age	Yes
20870	Single	Female	$10,000.00	2	High Schoo	Manual	Yes	1	0-1 Miles	Europe	38	Middle Age	Yes
23316	Single	Male	$30,000.00	3	Partial Col	Clerical	No	2	1-2 Miles	Pacific	59	Old	Yes
12610	Married	Female	$30,000.00	1	Bachelors	Clerical	Yes	0	0-1 Miles	Europe	47	Middle Age	No
27183	Single	Male	$40,000.00	2	Partial Col	Clerical	Yes	1	1-2 Miles	Europe	35	Middle Age	Yes
25940	Single	Male	$20,000.00	2	Partial Hig	Clerical	Yes	2	5-10 Miles	Pacific	55	Old	Yes
25598	Married	Female	$40,000.00	0	Graduate I	Clerical	Yes	0	0-1 Miles	Europe	36	Middle Age	Yes
21564	Single	Female	$80,000.00	0	Bachelors	Professional	Yes	4	More than	Pacific	35	Middle Age	No
19193	Single	Male	$40,000.00	2	Partial Col	Clerical	Yes	0	1-2 Miles	Europe	35	Middle Age	Yes
26412	Married	Female	$80,000.00	5	High Schoo	Management	No	3	5-10 Miles	Europe	56	Old	No

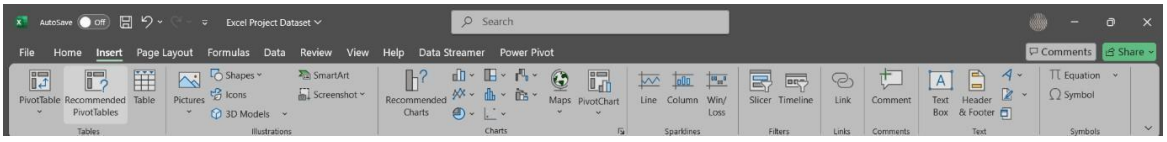
Step 4: For the different group age peoples we using the formula [=IF(L2>54,"Old",IF(L2>=31,"Middle Age",IF(L2<31,"Adolescent","Invalid")))]



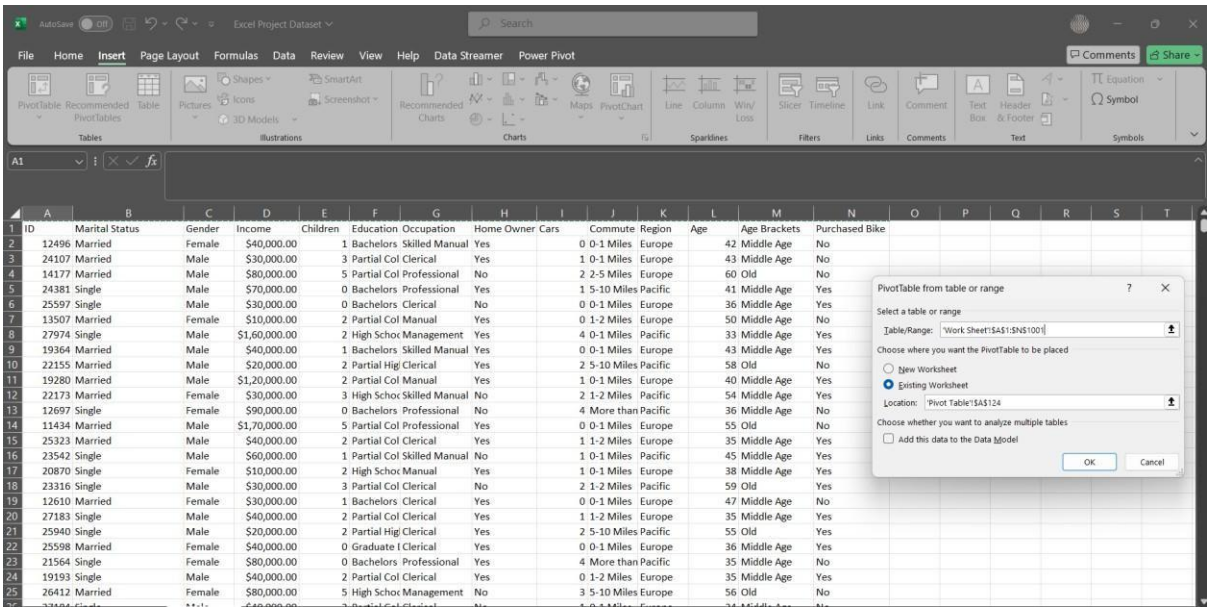
=M28:M32

26412	Married	Female	$80,000.00	5	High Schoo	Management	No	3	5-10 Miles	Europe	56	Old	No
27184	Single	Male	$40,000.00	2	Partial Col	Clerical	No	1	0-1 Miles	Europe	34	Middle Age	No
12590	Single	Male	$30,000.00	1	Bachelors	Clerical	Yes	0	0-1 Miles	Europe	63	Old	No
17841	Single	Male	$30,000.00	0	Partial Col	Clerical	No	1	0-1 Miles	Europe	29	Adolescent	Yes
18283	Single	Female	$1,00,000.00	0	Bachelors	Professional	No	1	5-10 Miles	Pacific	40	Middle Age	No
18299	Married	Male	$70,000.00	5	Partial Col	Skilled Manual	Yes	2	5-10 Miles	Pacific	44	Middle Age	No
16466	Single	Female	$20,000.00	0	Partial Hig	Manual	No	2	0-1 Miles	Europe	32	Middle Age	Yes
19273	Married	Female	$20,000.00	2	Partial Col	Manual	Yes	0	0-1 Miles	Europe	63	Old	No
22400	Married	Male	$10,000.00	0	Partial Col	Manual	No	1	0-1 Miles	Pacific	26	Adolescent	Yes
20942	Single	Female	$20,000.00	0	High Schoo	Manual	No	1	5-10 Miles	Europe	31	Middle Age	No
18484	Single	Male	$80,000.00	2	High Schoo	Skilled Manual	No	2	1-2 Miles	Pacific	50	Middle Age	Yes
12291	Single	Male	$90,000.00	5	Partial Col	Professional	No	2	2-5 Miles	Europe	62	Old	Yes
28380	Single	Female	$10,000.00	5	Partial Hig	Manual	No	2	0-1 Miles	Europe	41	Middle Age	No
17891	Married	Female	$10,000.00	2	Partial Col	Manual	Yes	1	0-1 Miles	Europe	50	Middle Age	Yes
27832	Single	Female	$30,000.00	0	Partial Col	Clerical	No	1	2-5 Miles	Europe	30	Adolescent	No
26863	Single	Male	$20,000.00	0	High Schoo	Manual	No	1	2-5 Miles	Europe	28	Adolescent	No
16259	Single	Female	$10,000.00	4	Partial Hig	Manual	Yes	2	0-1 Miles	Europe	40	Middle Age	Yes
27803	Single	Female	$30,000.00	2	Partial Col	Clerical	No	0	0-1 Miles	Europe	43	Middle Age	No
14347	Single	Female	$40,000.00	2	Bachelors	Management	Yes	2	5-10 Miles	Pacific	65	Old	Yes
17703	Married	Female	$10,000.00	1	Graduate I	Manual	Yes	0	0-1 Miles	Europe	40	Middle Age	No
17185	Married	Female	$1,70,000.00	4	Partial Col	Professional	No	3	5-10 Miles	Europe	48	Middle Age	Yes
29380	Married	Female	$20,000.00	3	High Schoo	Manual	Yes	0	0-1 Miles	Europe	41	Middle Age	Yes
23986	Married	Female	$20,000.00	1	Bachelors	Clerical	Yes	0	0-1 Miles	Europe	66	Old	Yes
24466	Married	Female	$60,000.00	1	Partial Col	Skilled Manual	Yes	1	5-10 Miles	Pacific	46	Middle Age	Yes
29097	Single	Female	$40,000.00	2	Partial Col	Skilled Manual	Yes	2	5-10 Miles	Pacific	52	Middle Age	Yes

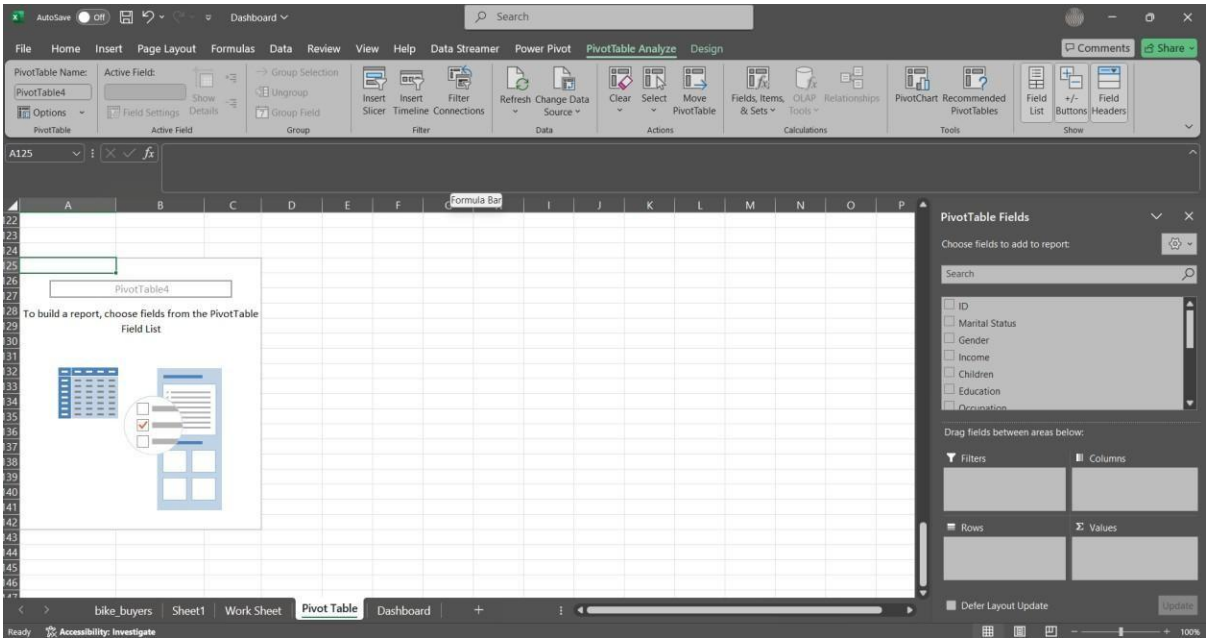
Step 5: After the completion of the data cleaning and pre-processing now we can start to create the pivot table. Make an add of a sheet in the same board to work pivot table in different table.



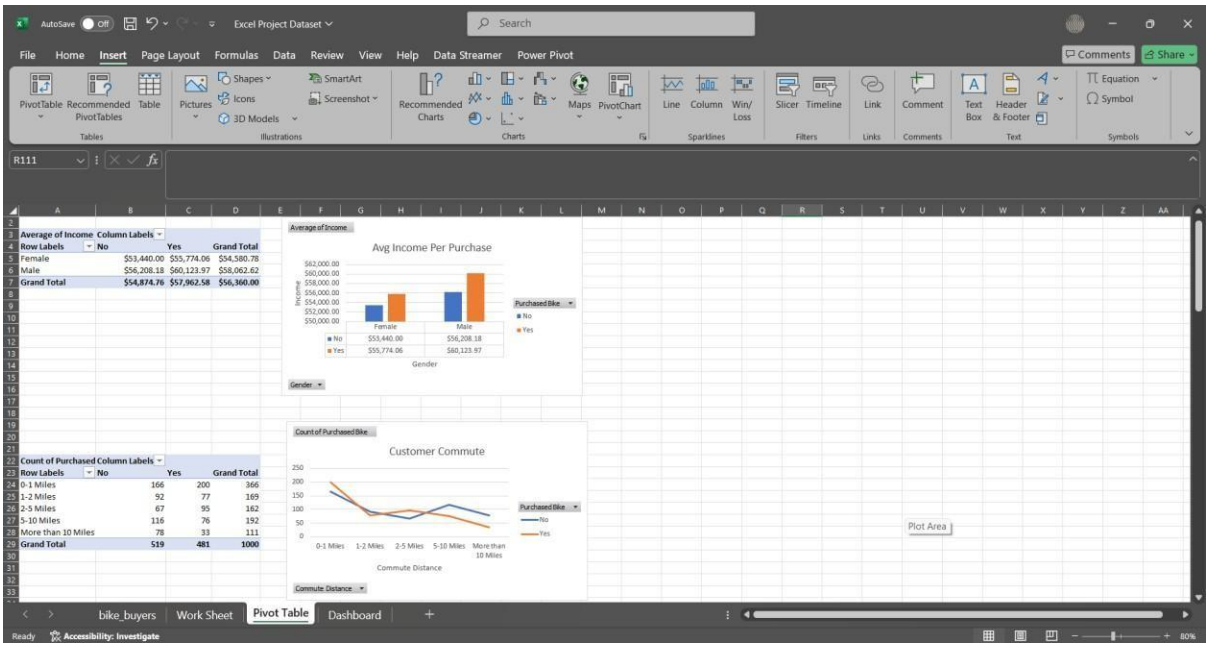
Step 6: Select the pivot table and in the down, you can see the dataset select that sheet and give ctrl+a to work on the pivot table.

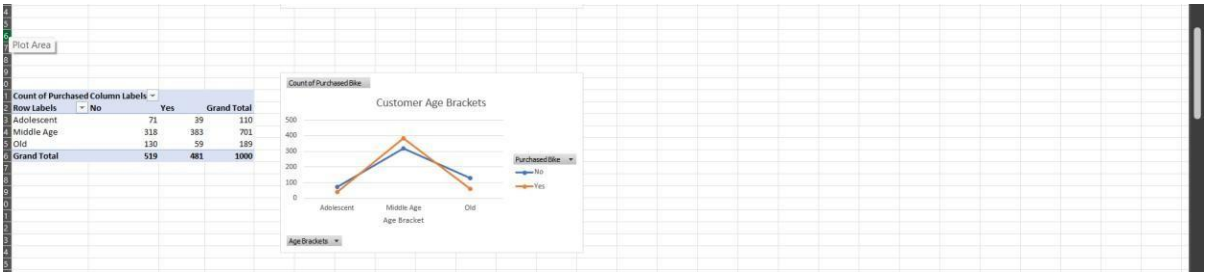


Step 7: For the user needs we can select and drag the option to make chart with following datasets to make understand the user we can use charts too.

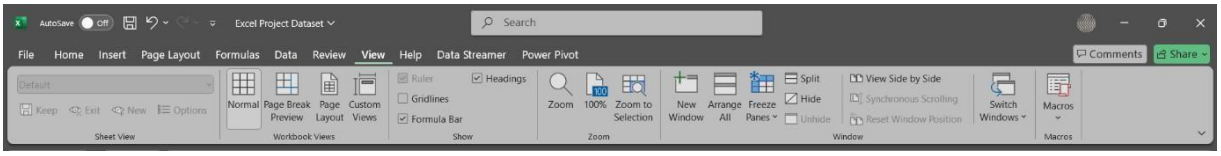


Step 8: Using recommended charts we can visualize the total value which we taken in pivot table.

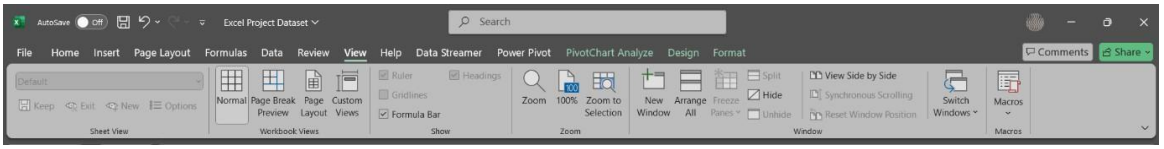




Step 9: Now creating a Dashboard in the excel sheet same like pivot table add a sheet for dashboard. Using the view option you can edit the grid lines.



Step 10: Design the dashboard to look with informative and user friendly. Using pivot chart analysis option, you can work on the design and slicer to know about the detail information of pivot chart.



RESULT:

Thus the Data Analysis and Visualization using Advanced Excel was implemented successfully.